UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL
INSTITUTO DE MATEMÁTICA E ESTATÍSTICA

REAL ANALYSIS: A LONG-FORM MATHEMATICS TEXTBOOK BY JAY CUMMINGS

STUDENT'S SOLUTIONS TO SELECTED EXERCISES

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1 The Reals

2 Cardinality

3 Sequences

4 Series

Exercise 4.3. Prove Proposition 4.18. That is, prove that if $\sum_{k=1}^{\infty} |a_k|$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.

Exercise 4.13.

(a) Prove that if (ka_k) converges to a nonzero number L, then the series $\sum_{k=1}^{\infty} a_k$ diverges. Give an example to show that the converse is false.

(b) Prove that if $(k^2 a_k)$ converges (to any real number), then the series $\sum_{k=1}^{\infty} a_k$ converges. Give an example to show that the converse is false.

Exercise 4.14. Prove that if $a_k > 0$ for all k and $\sum_{k=1}^{\infty} a_k^2$ converges, then $\sum_{k=1}^{\infty} \frac{a_k}{k}$ converges.

Exercise 4.15. Prove the *Cauchy condensation test*. That is, suppose that (a_k) is a decreasing sequence for which $a_k \to 0$. Prove that $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Exercise 4.17. Prove the *ratio test* via the following steps. Given a series $\sum_{k=1}^{\infty} a_k$ with $a_k \geq 0$ for all k, assume that

$$\lim_{k\to\infty} \frac{a_{k+1}}{a_k} =: r < 1.$$

We will show that $\sum_{k=1}^{\infty} a_k$ converges.

(a) Let q be such that $r < q < 1$. Explain why there is some N such that $n \geq N$ implies $|a_{k+1}| \leq |a_k| q$.

(b) Explain why $\sum_{k=1}^{\infty} |a_N| q^k$ necessarily converges.

(c) Finally, use part (b) to prove that $\sum_{k=1}^{\infty} |a_k|$ converges.

Proof. Let $\sum_{k=1}^{\infty} a_k$ be a series with $a_k \geq 0$ for all k. Assume that $\lim_{k\to\infty} \frac{a_{k+1}}{a_k} = r < 1$. Let q be such that $r < q < 1$. By the definition of sequential limits, there exists some N such that for all $n \geq N$, we have

$$\left| \left| \frac{a_{n+1}}{a_n} \right| - r \right| < \epsilon \qquad \forall \epsilon > 0. \tag{1}$$

or, letting $\epsilon = q - r$ (which is positive since $q > r$),

$$r - (q - r) < \left| \frac{a_{n+1}}{a_n} \right| < r + (q - r) \tag{2}$$

$$\left| \frac{a_{n+1}}{a_n} \right| < q \tag{3}$$

$$|a_{n+1}| < q\,|a_n|\,. \tag{4}$$

Exercise 4.18. Prove the *root test* via the following steps. Given a series $\sum_{k=1}^{\infty} a_k$ where each $a_k \geq 0$, assume that the limit $\lim_{k\to\infty} (a_k)^{1/k}$ exists. Call this limit ρ. Then this series converges if $\rho < 1$ and diverges if $\rho > 1$. If $\rho = 1$, then the test is inconclusive.

(a) Suppose $\rho < 1$. Let $\epsilon = \frac{1-\rho}{2}$ and $\rho_1 = \rho + \epsilon$. Prove that there is some N for which $(a_n)^{1/n} < \rho_1$ for all $n \geq N$.

(b) Prove that $\sum_{k=N}^{\infty} a_k$ converges by comparing it to a geometric series. Then conclude that $\sum_{k=1}^{\infty} a_k$ also converges.

(c) Suppose $\rho > 1$. Let $\epsilon = \frac{\rho-1}{2}$ and $\rho_2 = \rho - \epsilon$. Prove that there is some N for which $(a_n)^{1/n} > \rho_2$ for all $n \geq N$.

(d) Use this to argue that $\sum_{k=N}^{\infty} a_k$ diverges by using the divergence test. Then conclude that $\sum_{k=1}^{\infty} a_k$ also diverges.

Proof. Let $\sum_{k=1}^{\infty} a_k$ be a series with $a_k \geq 0$ for all k. Assume that the limit $\lim_{k\to\infty}(a_k)^{1/k}$ exists and call this limit ρ. First, suppose $\rho < 1$ and let $\epsilon = \frac{1-\rho}{2}$. Then set $\rho_1 = \rho + \epsilon$. By the definition of sequential limits, there exists some N such that for all $n \geq N$, we have

$$|(a_n)^{1/n} - \rho| < \epsilon \qquad \forall \epsilon > 0. \tag{5}$$

or, for $\rho < 1$,

$$\rho + -\frac{1-\rho}{2} < (a_n)^{1/n} < \rho + \frac{1-\rho}{2} \tag{6}$$

$$(a_n)^{1/n} < \rho + \frac{1-\rho}{2} = \rho + \epsilon = \rho_1. \tag{7}$$

From (7), we have that $0 \leq a_k < p^k$. Note that $\sum_{k=1}^{\infty} p^k$ converges for $p < 1$, by the geometric series test. Thus, by the comparison test, we have that $\sum_{k=N}^{\infty} a_k$ converges. Since $\sum_{k=1}^{N-1} a_k$ is finite, we conclude that $\sum_{k=1}^{\infty} a_k$ converges.

Now, suppose $\rho > 1$. Let $\epsilon = \frac{\rho-1}{2}$ and set $\rho_2 = \rho - \epsilon$. Again, by the definition of sequential limits, there exists some N such that for all $n \geq N$, we have

$$|(a_n)^{1/n} - \rho| < \epsilon \qquad \forall \epsilon > 0. \tag{8}$$

or, for $\rho > 1$,

$$\rho - \frac{\rho-1}{2} < (a_n)^{1/n} < \rho + \frac{\rho-1}{2} \tag{9}$$

$$(a_n)^{1/n} > \rho - \frac{\rho-1}{2} = \rho - \epsilon = \rho_2. \tag{10}$$

From (10), we have that $a_k > p^k$ for k sufficiently large. Note that $\sum_{k=1}^{\infty} p^k$ diverges for $p > 1$, by the divergence test. Thus, by the comparison test, we have that $\sum_{k=N}^{\infty} a_k$ diverges. Since $\sum_{k=1}^{N-1} a_k$ is finite, we conclude that $\sum_{k=1}^{\infty} a_k$ diverges. ■

5 The Topology of $\mathbb{R}$

6 Continuity

7 Differentiation

8 Integration

9 Sequences and Series of Functions